cover. To rectify this disadvantage, inert gases are filled in this lamp. Presence of inert gas causes heat loss. To compensate the heat loss, the filament is made as coiled wire. Increase in length of the filament leads to an increase in power.

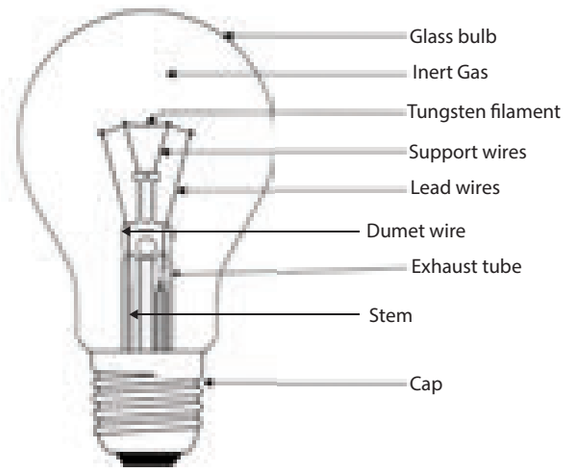


Fig. 2.5 Gas filled lamp

a. Working principle of a filament or an incandescent lamp

As we know, when a room heater is switched on, it gives out red light with heat at the working temperature of 750°C. At this temperature, the radiations are produced by infra red rays.

When an electric current is passed through a fine metallic wire, it raise the temperature of the wire. At low temperature, only heat is produced but at a higher temperature light radiation goes on increasing. The filament lamp consists of fine wire of high resistive material placed in an evacuated glass bulb. This type of lamp is operated at the temperature of 2500°C. A tungsten filament is covered in an evacuated glass bulb. But to improve the life of the filament, some chemicals like argon or nitrogen or neon gases are filled.

b. Properties of metal for filament

1. It can be operated at a high temperature, since it has a high melting point.
2. It produces more heat because it has a high specific resistance.
3. Filament resistance may not change at the operating temperature because it has a low temperature co-efficient.
4. Because of low vapour pressure, it may not get vapour.
5. Because of high ductility, it may withstand mechanical vibrations.

2.6 Sodium vapour lamp and mercury vapour lamp

2.6.1 Sodium vapour lamp

Sodium vapour lamps are some of the most efficient lamps in the world. They have an efficiency of up to 190 lumens per watt compared to an incandescent street lamp which has between 15 and 19 lumens per watt.

This sodium vapour lamps comes in two major groups:

1. High pressure sodium vapour lamps (HPS)
2. Low pressure sodium vapour lamps (LPS)

This lamp consists of discharge tube made from special heat resistance glass, containing a small amount of metallic sodium, neon gas and two electrodes. Neon gas is added to start the discharge and to develop enough heat to vapour sodium. A long tube is required to get more light from this lamp. To reduce overall dimensions of the lamp, the tube is generally bent into U-shape.

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